



KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN CIVIL ENGINEERING

ECV 210: CIVIL ENGINEERING MATERIALS II

2ND SEMESTER 2020/2021

TIME: 2 HOURS

INSTRUCTIONS: Answer question one and any other two questions.

QUESTION ONE (COMPULSORY)

- a) Discuss the application of steel in construction works. (4 marks)
- b) Where and why do we use timber in civil engineering projects (4 marks)
- c) Discuss the application of polymers in construction industry (4 marks)
- d) Define seasoning of timber and describe the methods of seasoning (4 marks)
- e) Discuss the types of defects in timber as a construction material (4 marks)
- f) Describe the use of ceramics in construction industry. What are their benefits over the other materials (4 marks)
- g) Discuss durability of Steel, timber and ceramics as civil engineering materials. (4 marks)
- h) Discuss the factors that determine the choice between timber and steel as construction materials. (2 marks)

QUESTION TWO

- A) Using well labeled diagram, discuss the manufacture process of steel (8 marks)
- B) Describe the mechanical and physical properties of reinforcement bars. (4 marks)
- C) Discuss corrosion of steel and the methods of preventing it. (4 marks)
- D) Define carbon steel and discuss the different types of reinforcement steel (4 marks)

QUESTION THREE

- A) Discuss the mechanical and physical properties of timber as a construction material (8 marks)
- B) Differentiate between hardwood and softwood. (4 marks)
- C) Discuss grading of timber and explain the methods of undertaking it. (4 marks)
- D) Discuss the factors that determine the compressive strength of timber (4 marks)

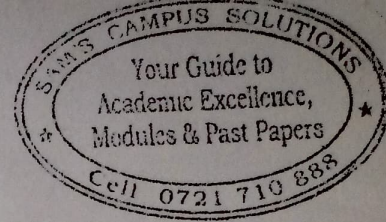
QUESTION FOUR

- A) Discuss the manufacturing process of ceramic tiles and describe the properties of ceramics that make suitable for construction works. (10 marks)
- B) Where and why do we use aluminum material in construction Industry?. What Properties of this material determine its choice over other materials. (5 marks)
- C) Discuss the manufacturing process of polymers used in construction (5 marks)

QUESTION FIVE

- A) Using well labeled diagrams, discuss the procedure for carrying out the tensile strength test for both steel and timber. (10 marks)
- B) Discuss the importance of composite products used in construction works and gives examples of these products. (6 marks)
- C) Describe the application of polymers in construction works. (4 marks)

ECV 210



Kenyatta University

Sit In CAT- 2nd Semester University Examinations 2013/14

Department of Civil Engineering

ECV 210- Civil Engineering Materials

2 Hours

Answer all Questions

QUESTION ONE

- i) 1) Discuss civil engineering materials ; 2 marks ✓
2) Provide two examples and properties of gypsum construction materials; 3 marks
- Plaster & Gypsum
- ii) 1) Briefly explain any typical materials testing ; 2 marks
2) Provide a typical classification of any materials testing- 3 marks.
- iii) Discuss two typical properties of concrete materials -5 marks
- iv) Explain why sampling and testing is critical of any material. 5 marks

QUESTION TWO

- 1) Briefly discuss ready mixed concrete as civil engineering material; 5 marks
- 2) Discuss two typical specification of bituminous material according to its use; 5 marks
- 3) Explain two examples of classification of stones; 4 marks
- 4) Explain design and construction of stone building; 6 marks

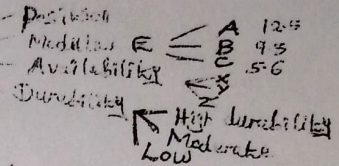
Geological
Physical
Chemical Properties
Stratification
Weathering
Foliation
Argillaceous
Siliceous
Calcareous

QUESTION THREE

- 1) Describe the constituents of plastics; 4 marks ✓
- 2) State methods used for the fabrication of commercial articles from plastics; 4 marks ✓
- 3) Explain compression moulding; 4 marks ✓
- 4) Explain injection moulding; 4 marks ✓
- 5) Explain transfer moulding; 4 marks ✓

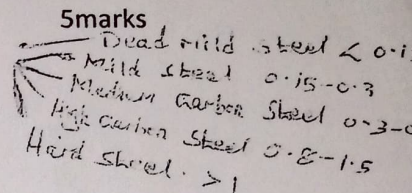
QUESTION FOUR

- 1) Discuss all the classifications of trees; 4 marks ✓
- 2) Explain structure of timber; 4 marks ✓
- 3) Explain the classification of timber; 4 marks ✓
- 4) Explain the seasoning of timber; 4 marks ✓
- 5) Define properties of wood fire resistance; 4 marks ✓



QUESTION FIVE

- I) Discuss the fundamental chemical principles in the extraction of iron from the ore. 5 marks
- II) Explain steel as the most suitable building /construction material among metallic materials. 5 marks
- III) Explain four types of steel on the basis of carbon content. 4 marks
- IV)
 - 1) Discuss six prominent steel-making processes; 2 marks
 - 2) Discuss properties and uses of mild steel; 2 marks
 - 3) Discuss factors influencing the properties of steel; 2 marks

Fire resistance

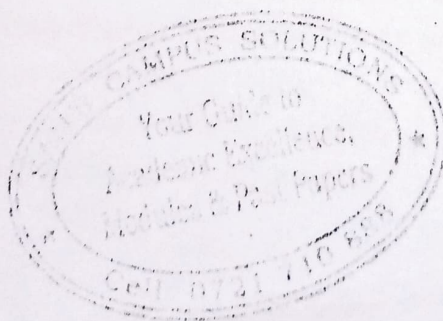
113°C → vapourisation of moisture

(110°C - 165°C) → Volatilisation of extraneous materials

(165°C - 220°C) → decomposition begins → evolution of inflammable gases → ignite and form first flames around heated wood.

(220°C - 370°C) → Wood begins to glow and ignite.

- Until this point is reached the wood portion does not support its own combustion.



Besse

L.D

Craci

Dupl

- 1. - Stirring
- Ladle furnace
- Ladle injection
- Degassing
- AS-OB

back
back
Rambium
Sap
Heard - firm support
Pith
Medullary